

90-DAY MODERN JAVA DEVELOPER ROADMAP (DAY-WISE PLAN)

PHASE 1: CORE JAVA + SPRING BOOT (DAYS 1–30)

WEEK 1 – Modern Java Foundations

Day 1: Java basics, data types, operators

Day 2: OOP—Encapsulation, Abstraction

Day 3: OOP—Inheritance, Polymorphism

Day 4: Collections Framework

Day 5: Generics + Enum

Day 6: Exception handling + custom exceptions

Day 7: 20 Java coding exercises

WEEK 2 – Advanced Java

Day 8: Threads, Runnable, Callable

Day 9: ExecutorService, ForkJoinPool

Day 10: CompletableFuture

Day 11: Synchronization, Locks

Day 12: JVM Internals

Day 13: Lambda, Streams, Optional

Day 14: Multi-threading project

WEEK 3 – Spring Boot Foundations

Day 15: Spring Boot basics, starters

Day 16: Controllers, Services, Repositories

Day 17: JPA + MySQL CRUD

Day 18: Relationships (OneToMany etc.)

Day 19: Pagination + Sorting

Day 20: Flyway migrations

Day 21: User/Product CRUD mini project

WEEK 4 – Advanced Spring Boot

Day 22: Profiles, Environment

Day 23: Exception Handling + ControllerAdvice

Day 24: AOP Logging

Day 25: Validation

Day 26: Redis caching

Day 27: Logging (SLF4J)

Day 28–30: Production-ready API build

PHASE 2: MICROSERVICES + KAFKA + CLOUD (DAYS 31–60)

WEEK 5 – Microservices

Day 31: Microservices concepts

Day 32: Feign Clients

Day 33: API Gateway

Day 34: Eureka Server/Clients

Day 35: Load Balancer

Day 36–37: Build Order/Payment/Inventory services

WEEK 6 – Kafka + Resilience4j

Day 38: Kafka basics

Day 39: Producers & Consumers

Day 40: Topics, Partitions, Offsets

Day 41: Acks, ISR, Delivery semantics

Day 42: Dead Letter Queue

Day 43: Retry & Circuit Breaker

Day 44: Kafka + Notification service

WEEK 7 – Config Server + Security

Day 45: Config Server setup

Day 46: Git-backed config

Day 47: Encrypt/Decrypt

Day 48: JWT Authentication

Day 49: JWT Authorization

Day 50–51: Secure microservices integration

WEEK 8 – DB + Testing

Day 52: JPA advanced

Day 53: Transactions

Day 54: TestContainers

Day 55: JUnit 5 unit tests

Day 56: Integration tests

Day 57–60: Banking/eCommerce module

PHASE 3: DEVOPS + K8S + SYSTEM DESIGN (DAYS 61–90)

WEEK 9 – Docker + Kubernetes

Day 61: Dockerfile

Day 62: Multi-stage builds

Day 63: Docker Compose

Day 64: K8s basics

Day 65: Deploy Spring Boot to Minikube

Day 66: ConfigMap/Secrets

Day 67–69: K8s microservices

WEEK 10 – CI/CD + CLOUD

Day 70: GitHub Actions pipeline

Day 71: Build + Test stages

Day 72: Docker image automation

Day 73: AWS EC2

Day 74: AWS RDS

Day 75–77: Cloud deployment

WEEK 11 – Observability

Day 78: ELK Stack

Day 79: Prometheus

Day 80: Grafana Dashboards

Day 81–84: Monitoring setup

WEEK 12 – System Design + Interview Prep

Day 85: API design + scaling

Day 86: Caching + databases

Day 87: Load balancers

Day 88: HLD examples (Uber/Netflix)

Day 89: LLD examples

Day 90: Final revision + project review